Machine Learning Algorithm Exam Cheat Sheet

Quick identification guide for case-study/open-book exams

Question Keyword / Output	Use This Algorithm
Spam, Ham, Email, SMS, Review, Text	Naive Bayes / Logistic Regression + TF-IDF or CountVectorizer
Nearest Similar Records	KNN
IF-ELSE Rules	Decision Tree
Many Decision Trees	Random Forest
Correct Previous Errors	Gradient Boosting
Advanced Boosting / Best Accuracy	XGBoost
Category Prediction (Yes/No)	Logistic Regression or SVC
House Price, Salary, Sales	Linear Regression
Many Inputs -> One Number	Multiple Linear Regression
Curve / Non-linear Trend	Polynomial Regression
Complex Numerical Prediction	SVR
Flower Classification	Naive Bayes
Customer Churn	Logistic Regression / Random Forest / XGBoost

10-Second Rule

Output is a Category (Yes/No, Spam/Ham, Pass/Fail) -> Classification. Use Logistic Regression, Naive Bayes, KNN, Decision Tree, Random Forest, SVC, XGBoost.
Output is a Number (Price, Salary, Sales, Charges) -> Regression. Use Linear Regression, Multiple Linear Regression, Polynomial Regression, SVR.